

NAVIDU LAKSHAN MADUSHANKA

Software Engineering Intern

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PROFESSIONAL SUMMARY

Final year Computer Engineering undergraduate with a fundamentals first background in full stack web development, REST API design and software quality practices, built on strong data structures, algorithms and object oriented design foundations. Hands on experience across five full stack, real time and machine learning projects, including live production deployments and a multi module Java REST API with an automated JUnit 5, Checkstyle and JaCoCo quality pipeline, working in Agile and Scrum teams to turn stakeholder requirements into working features. Fast, self directed learner currently extending this foundation into Python, Golang, Terraform and GCP.

CORE COMPETENCIES

Full Stack Web Development | Software Engineering | REST API Design & Development | Object Oriented Programming (OOP) | Data Structures & Algorithms | Software Quality Assurance & Testing | Agile and Scrum Methodology | CI/CD & DevOps Practices | Database Design (SQL and NoSQL) | Version Control (Git and GitHub) | Machine Learning Fundamentals | Technical Documentation | Cross Functional Team Collaboration

EDUCATION

University of Jaffna Jan 2023 - Present

BSc (Hons) in Engineering, Computer Engineering (Undergraduate)

Relevant Coursework: Software Engineering, Object Oriented Programming, Database Management Systems, Computer Networks & Security, Operating Systems, Computer Architecture, Embedded Systems Design, Data Structures & Algorithms, Machine Learning, Deep Learning, Artificial Intelligence, Advanced Networking.

Bandaranayake Central College, Veyangoda 2020

G.C.E. Advanced Level, Physical Science Stream

Results: B in Combined Mathematics, B in Chemistry, B in Physics

RELEVANT SKILLS

- **Programming & CS Fundamentals:** Java (OOP, REST APIs), JavaScript, TypeScript, Python, HTML, CSS, PHP, Data Structures & Algorithms, object oriented design
- **Web & API Development:** React.js, Node.js, Express.js, Next.js, REST API design and architecture, API integration
- **Software Quality & Testing:** JUnit 5, Checkstyle, JaCoCo, unit and integration testing, code review practices, automated CI pipelines (GitHub Actions)
- **Databases:** MySQL, MongoDB, relationships and ongoing maintenance
- **Machine Learning & Data:** Python, pandas, scikit learn
- **Networking & Security:** JWT, RBAC authentication, WebRTC, Socket.io, application security fundamentals
- **Tools & Practices:** Git, GitHub, Docker, CI/CD, Postman, VS Code, Linux, Agile, Scrum, Jira, Software Development Life Cycle (SDLC), technical documentation, problem solving and debugging
- **Expanding Tech Stack (Learning):** Golang, C++, GCP, Terraform, backed by a strong OOP and full stack foundation
- **Non Technical Skills:** Communication & Stakeholder Collaboration, Teamwork, Problem-Solving & Analytical Thinking, Adaptability, Time Management, Leadership

CERTIFICATIONS & COURSES

- **IBM DevOps and Software Engineering Professional Certificate**, Coursera, IBM (13-course program: DevOps, Agile and Scrum, Git, Linux, cloud, containers, security, Python and Flask) ([Verify](#))
- **Software Engineering: Modeling Software Systems using UML**, Coursera, HKUST ([Verify](#))
- **DevOps Pro: Linux, Docker, Kubernetes, CI/CD & IaC Specialization**, Coursera, KodeKloud ([Verify](#))
- **DevOps Complete Course Specialization**, Coursera, Packt ([Verify](#))
- **IBM DevOps, Cloud, and Agile Foundations Specialization**, Coursera, IBM ([Verify](#))

FEATURED PROJECTS

DepositLK (SaveSmart): Full Stack Financial Advisory Platform | Solo Project | Mar 2025 | [Link](#)

Tech Stack: Next.js, TypeScript, React, Prisma, Docker, CI/CD

- Set out to independently deliver a production ready financial advisory platform end to end, designing, building and shipping the full application as a solo project.
- Built the frontend with Next.js, TypeScript and React, backed by Prisma for data modeling, to give users a fast, type safe interface for advisory data.

- Automated builds and deployments with a Docker based CI/CD pipeline, enabling faster, more consistent releases.

Timelyx: Lecture Hall Management System / Team Project | Dec 2025 | [Link](#)

Tech Stack: React.js, Node.js, Express.js, MongoDB, JWT, Jira, Scrum

- The lecture hall booking process needed a centralized system for four distinct user roles, so our team built a full stack MERN application with role based dashboards for each.
- Implemented JWT based authentication and role-based access control to keep each user type restricted to its own permissions and views.
- Worked through weekly Agile and Scrum sprints in Jira, translating stakeholder requirements directly into delivered features on schedule.

Remote jam webrtc: Real Time Peer to Peer Platform / Solo Project | Dec 2025 | [Link](#)

Tech Stack: WebRTC, Socket.io, Node.js, JavaScript

- To enable direct, low latency audio and video communication between users, built a real time peer to peer platform using WebRTC.
- Implemented Socket.io for signaling and connection handling, adding auto reconnect support to keep sessions stable on unreliable networks.
- Backed the responsive frontend with a Node.js server for session management, ensuring smooth handoffs as users joined and left calls.

Laptop Price Predictor / Solo Project | Jun 2025 | [Link](#)

Tech Stack: Python, pandas, scikit-learn

- To help buyers estimate a fair laptop price from its specifications, built a machine learning model that predicts price from technical features.
- Cleaned and engineered the dataset with pandas to improve model input quality before training.
- Evaluated multiple algorithms with scikit learn and selected the best performing model based on accuracy.

Ticket Booking Application / Solo Project | Dec 2025 | [Link](#)

Tech Stack: React 19, Vite, Tailwind CSS, Bootstrap, Clerk, React Router

- As part of a frontend architecture research initiative, built a responsive ticket booking web application with React and Vite to evaluate a modern frontend stack.
- Implemented secure user authentication and sign up flows using Clerk, reducing the need for custom auth code.
- Designed a fully responsive UI with Tailwind CSS and Bootstrap and client side routing via React Router, giving users a smooth, single page booking experience.

DecodeLabs Backend API / Industrial Training Project | [Link](#)

Tech Stack: Node.js, Express.js, JWT, RBAC

- During a structured industrial training program, built a production style REST API with Node.js and Express to practice real world backend engineering standards.
- Implemented authentication, role-based access control and input validation across all endpoints, hardening the API against unauthorized access and bad input.

NexaFlow / Solo Project | [Link](#)

Tech Stack: HTML5, CSS3, JavaScript

- Set out to build an accessible, fully responsive site from scratch, creating a complete design system aligned with WCAG guidelines.
- Implemented a dark and light theme toggle with persistent preference handling, improving usability across different viewing conditions.

ADDITIONAL PROGRAMS & LEADERSHIP

- DecodeLabs Batch 2026: completed a structured series of full stack development projects.
- SpiritX Stage 1 (2024): participated in a 24 hour web development hackathon, delivering a working product under time pressure as part of a team.
- Editor, IEEE RAS Student Branch Chapter, University of Jaffna (Feb 2024 to Jan 2025): contributed to chapter communications and stayed engaged with the broader engineering and robotics community.

REFERENCES

Dr. (Mrs.) J. Jananie

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